A Floer-Theoretic Framework for Hamiltonian Bifurcations

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Contents

Based on a joint work with Hong-Kwon Jo (SNU), arXiv:2607.22178.

Main result. We describe the explicit changes of the local Floer chain complex for generic types of Hamiltonian bifurcations.

- **Bifurcation and Floer Homology.**
  Classification of bifurcations, relation with Floer homology.
- **Main Results.**
  Explicit changes of the local Floer chain complex across bifurcations.
- **Research Plans.**
  Toward more bifurcations, Floer-type Cerf theory.

Bifurcation and Floer Homology

Motivation. Bifurcations and Conley–Zehnder indices

A **bifurcation** is, heuristically, the birth or death of periodic orbits.

Several recent works [AFvK+23], [AB25], [JKvK26] relate bifurcations to symplectic invariants, for example Conley–Zehnder indices.

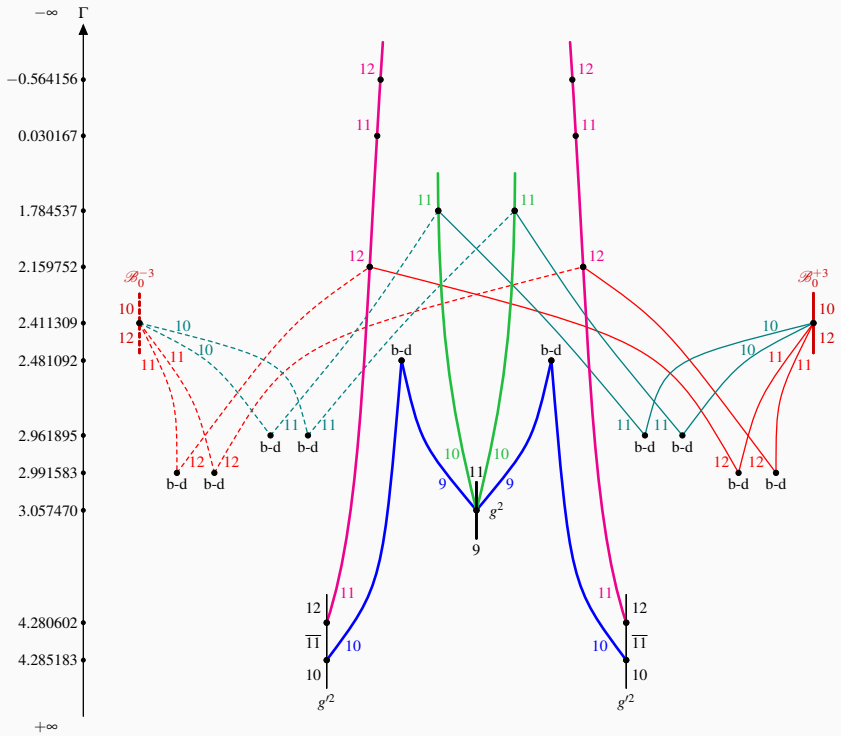


Figure 1: Bifurcation graph of Hill's lunar problem, [AB25].

Main Question

Bifurcation theory studies a change of structure of periodic orbits.

Floer theory studies a symplectic-topological invariant constructed by a Floer chain complex, which is generated by periodic orbits.

$\Rightarrow$ A bifurcation gives a **mutation** of a Floer chain complex, while the homology remains the same.

To study bifurcation through Floer theory, we need *chain-level analysis*.

Can we describe this effect explicitly?

Bifurcation

Target. 4-dimensional autonomous Hamiltonian system (W, ω, H).

A **Poincaré section** is a transversal section $\Sigma \subset H^{-1}(c)$ at p.

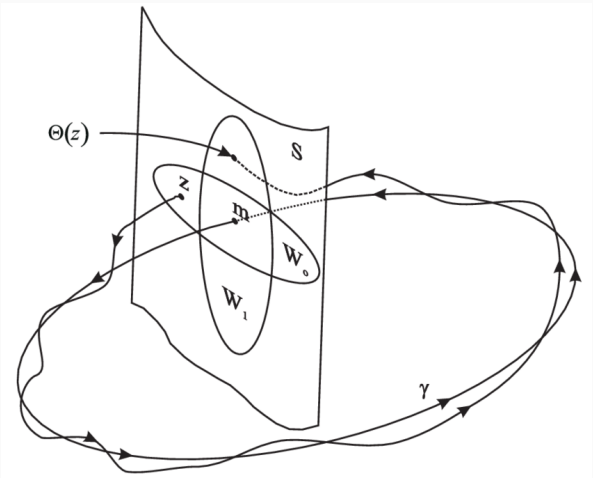


Figure 2: Poincaré section [OR99].

A **return map** is defined by $\Phi : U \subset \Sigma \to \Sigma$ by

$$\Phi(q) = Fl^{X_H}_{\tau(q)}(q), \quad \tau(q) = \min\left\{t \,:\, Fl^{X_H}_t(q) \in \Sigma\right\}.$$

Floquet multipliers are eigenvalues of $d_p\Phi$.

Idea. Periodic points of the return map give the periodic orbits.

Bifurcation

Let $H_\varepsilon : W^4 \to \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0.

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

We follow γ_ε and detect λ_ε. If $\lambda_\varepsilon = 1$ at some ε, then γ_ε is degenerate, and the orbit cylinder may break:

- periodic orbits may be born or vanish; this is called a **bifurcation**,
- if $\lambda_\varepsilon \neq 1$ but $\lambda_\varepsilon^k = 1$, then the k-th cover bifurcates.

Note. Conley–Zehnder index changes only when the orbit degenerates, and this fact is useful in practical purposes. ([AB25], [JKvK26].)

Example. Period Doubling

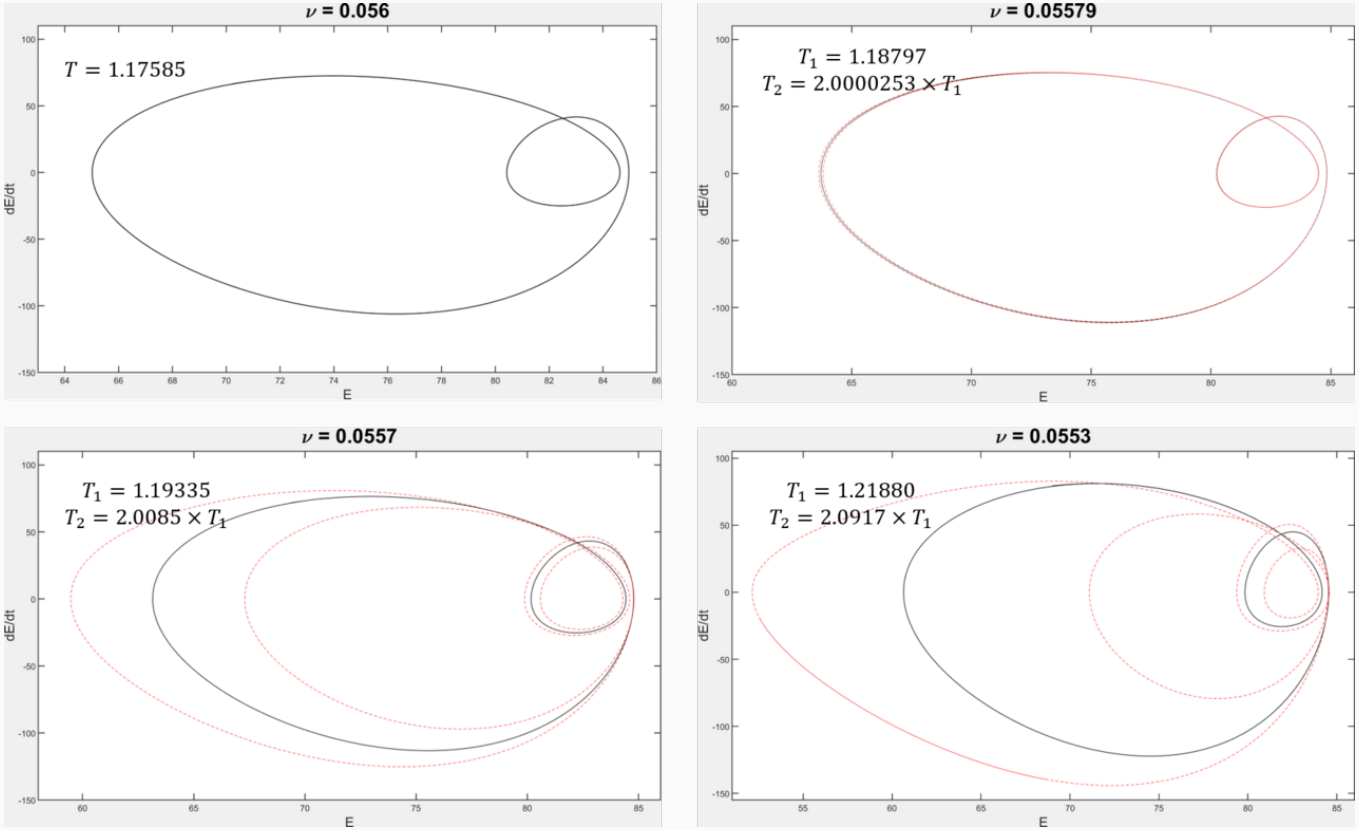


Figure 3: Period doubling in Kuramoto–Sivashinsky equation [Max21].

The red dashed orbit is the new orbit that emerges from the bifurcation.

This is a generic bifurcation of a double cover.

Floer Homology

Floer homology $HF_*(W, H)$ is a topological invariant associated with a symplectic manifold W and Hamiltonian H. (cf. [AD14].)

This is determined by the **Floer chain complex** $CF_*(W, H)$, whose

- generators are 1-periodic orbits of H,
- grading is given by CZ indices,
- the differential is determined by counting **Floer cylinder**, whose boundaries are asymptotic to periodic orbits.

We will use **local Floer homology**, which is a Floer homology on a neighborhood of a periodic orbit, $HF_*^{\mathrm{loc}}(\gamma_0)$.

Across a bifurcation, CF_* changes substantially;

- orbits are born or vanish $\Rightarrow$ the number of generators changes.
- the CZ index changes $\Rightarrow$ grading changes

However, the homology doesn't change, so the differential is responsible for keeping the homology invariant. We call this **mutation** of LFCC.

Main Results

Outline

We're interested in the case such that $\lambda^k = 1$.

1. By **Birkhoff normal form**, find a generating Hamiltonian G_ε of Φ^k.
2. Construct **Model Hamiltonian** H_ε from G_ε.
3. Using **Floer cascade** to compute Floer differential.

Birkhoff Normal Form

Key Idea. For generic Hamiltonian H_ε, if $\lambda_0^k = 1$, there exists an *autonomous* Hamiltonian G_ε, consists of $\mathbb{Z}_k$-symmetric polynomials, which generates k-th iterate of the Poincaré return map Φ^k.

$$G_\varepsilon = \begin{cases} \alpha u^2 + \beta v^3 + \varepsilon(au + bv + \ldots) & k = 1, \\ \alpha u^2 + \beta v^4 + \varepsilon(av^2 + \ldots) & k = 2, \\ \alpha \rho^4 + \beta \rho^k \cos k\varphi + \varepsilon(a\rho^2 + \ldots) & k \geq 3. \end{cases}$$

Critical points of G_ε corresponds to the k-periodic orbit of H_ε.

- In 1970, Meyer [Mey70] classified generic bifurcations using Birkhoff normal form of the Poincaré return map.
- This is similar to the resolution of singularity.

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70], which means the non-zero leading coefficients.

They are organized by the covering number k of the underlying orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Note. The fourth cover is a borderline case, depending on the coefficients of the normal form.

Model Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian of Φ_ε^k.

We construct a model Hamiltonian on $I_r \times (\mathbb{R}/k\mathbb{Z})_\theta \times \Sigma_z$ by,

$$H_\varepsilon = kh(r) + G_\varepsilon(z).$$

where $h'(0) = 1$ and $h''(0) \neq 0$.

- k-to-1 correspondence between $\mathrm{Crit}(G_\varepsilon)$ and $\mathrm{Per}(H_\varepsilon)$, say $p \leftrightarrow \gamma_p$.
- $\mu_{CZ}(\gamma_p) = 1 - \mathrm{Ind}_{\mathrm{Morse}}(p)$.
- A gradient flow between p and q gives a Floer cylinder between γ_p and γ_q, which we call a **gradient revolution**.

Note. H_ε and $\tilde{H}_\varepsilon$ are different, but has the same return map, bifurcation type, and Floer-type data up to finite degree and symplectomorphism.

Floer Cascade

Our system is autonomous, so we need to consider Morse–Bott type cascades to compute the differential between $\hat{\gamma}$ (max) and $\check{\gamma}$ (min).

Floer cascade is a sequence of Morse trajectories on the orbit and Floer cylinders between orbits.

If there is a Floer cylinder $u(s,t)$ from γ_+ to γ_-, for $p \in \gamma_\mp$, we define

$$\mathrm{ev}^{\pm}(p) = \left\{ q \in \gamma_\pm \ : \text{ for } t_0 \text{ such that } p = \lim_{s \to \mp\infty} u(s,t),\ q = \lim_{s \to \pm\infty} u(s,t) \right\}.$$

1. $\mu(\gamma_+;\gamma_-) = 1$, $\hat{\gamma}_+ \to \hat{\gamma}_-$.
   (Morse from $\hat{\gamma}_+$ to $\mathrm{ev}^+(\hat{\gamma_-})$, Floer cylinder from γ_+ to γ_-)
2. $\mu(\gamma_+;\gamma_-) = 1$, $\check{\gamma}_+ \to \check{\gamma}_-$.
   (Floer cylinder from γ_+ to γ_-, Morse from $\mathrm{ev}^-(\check{\gamma}_+)$ to $\check{\gamma_-}$)
3. $\mu(\gamma_+;\gamma_-) = 2$, $\check{\gamma}_+ \to \hat{\gamma}_-$.
   (Floer cylinder from γ_+ to γ_0, Morse from $\mathrm{ev}^-(\check{\gamma}_+)$ to $\mathrm{ev}^+(\hat{\gamma}_-)$, Floer cylinder from γ_0 to γ_-)

Floer Cascade

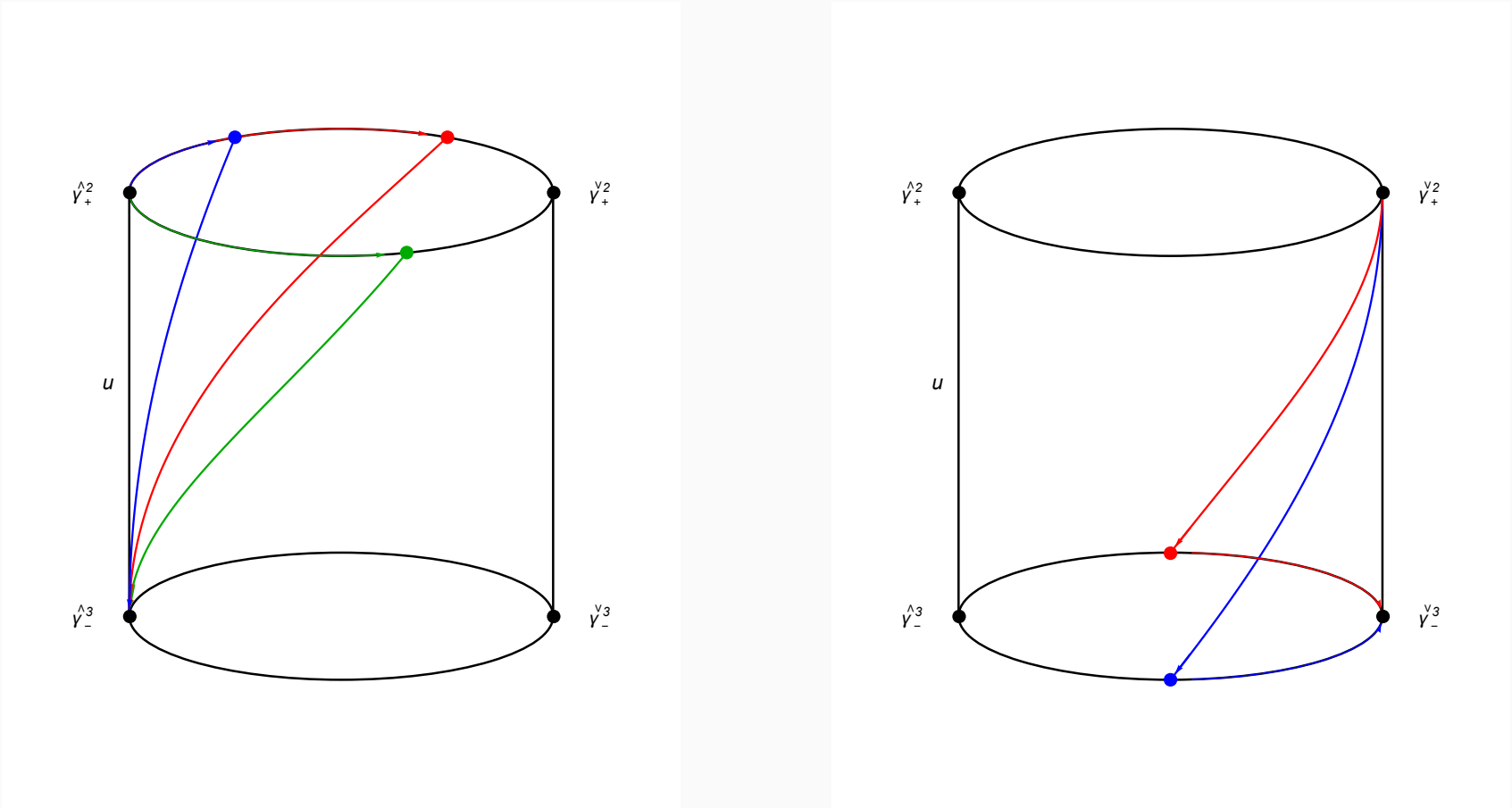


Figure 4: Cascades between 2nd and 3rd covers.

Cascade Counting Theorem

Theorem

Let $\gamma_\pm$ be two non-constant $1/k_\pm$-periodic orbits, so that $\gamma_\pm^{k_\pm}$ are 1-periodic orbits. Moreover, assume that

$$\mu_{CZ}(\gamma_+^{k_+}) - \mu_{CZ}(\gamma_-^{k_-}) = 1.$$

Then, the quotient $\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1$ is a 0-dimensional $\mathbb{Z}_{(k_+,k_-)}$-orbifold, and we have

$$\begin{aligned}
\#_2\mathcal{M}(\hat{\gamma}_-^{k_-}, \hat{\gamma}_+^{k_+}) &\equiv k_- \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod 2, \\
\#_2\mathcal{M}(\check{\gamma}_-^{k_-}, \check{\gamma}_+^{k_+}) &\equiv k_+ \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod 2.
\end{aligned}$$

In particular, if there is a cylinder from a simple γ_0 to a multiple cover γ_1^k,

$$\partial\hat{\gamma}_0 = k\hat{\gamma}_1^k, \quad \partial\check{\gamma}_0 = \check{\gamma}_1^k.$$

Birth-death $(k = 1)$

A generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 + \ldots + \varepsilon(u + \ldots).$$

- $(\varepsilon < 0)$ There are no periodic orbits.
- $(\varepsilon > 0)$ One elliptic γ_0 and one hyperbolic orbit γ_1 are born.
- There is one gradient trajectory between them.

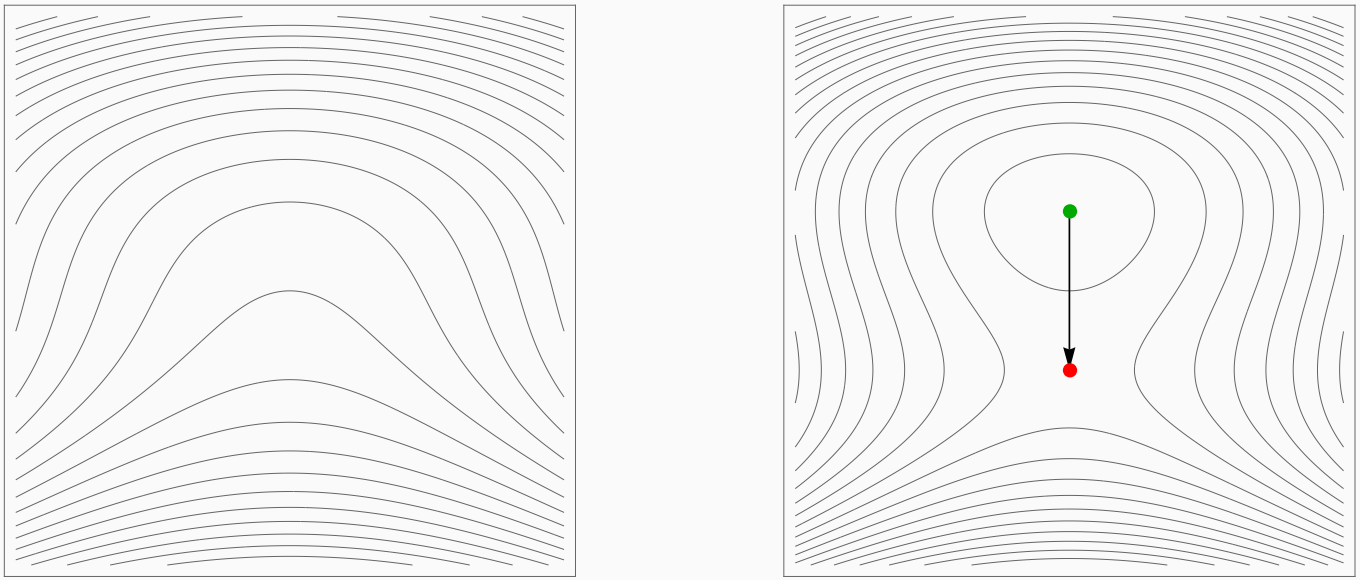


Figure 5: Critical points and gradient trajectories of birth-death bifurcation

Birth-death $(k = 1)$

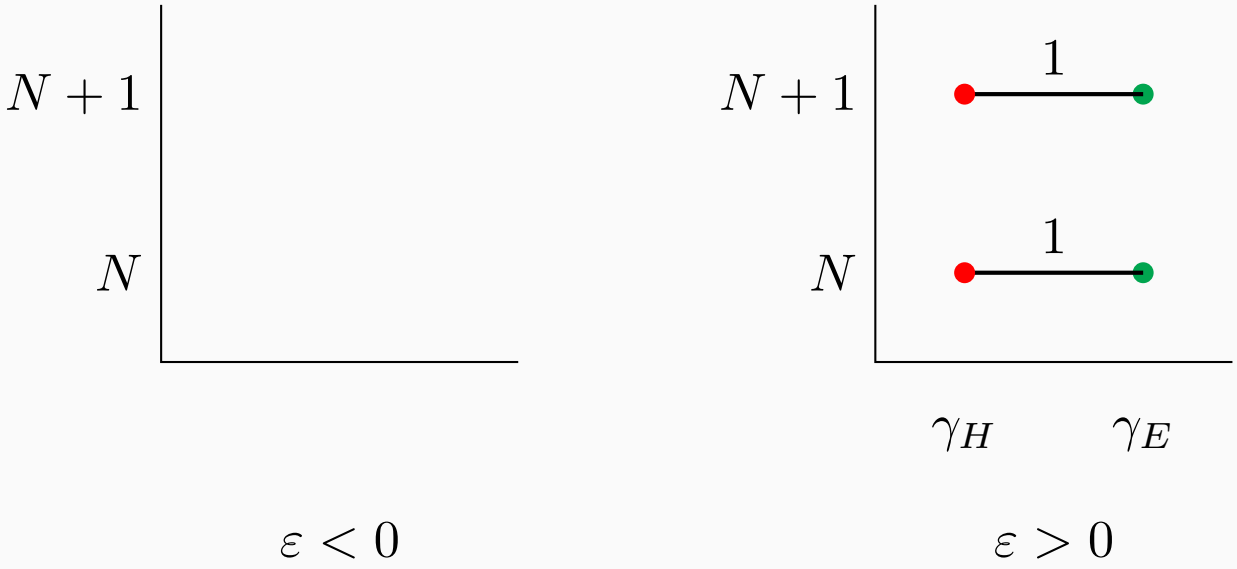


Figure 6: Chain complex mutation across the birth-death.

- Points at (n, m) corresponds to a generator of degree $n + m$.
- The line indicates the differential; $\partial(\hat{\gamma}_E) = \hat{\gamma}_H$ here.
- We are using $\mathbb{Z}_2$-coefficients.

We have $HF_*^{\mathrm{loc}}(\gamma) = 0$, where γ is the degenerate orbit at $\varepsilon = 0$.

Period doubling $(k = 2)$

A generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 + \ldots + \varepsilon(u^2 + \ldots).$$

- $(\varepsilon < 0)$ One elliptic orbit γ_0^2 exists.
- $(\varepsilon > 0)$ γ_0^2 becomes hyperbolic, and an elliptic orbit $(\bullet)$ γ_{new} is born.
- There is one cylinder between γ_0^2 and γ_{new}.

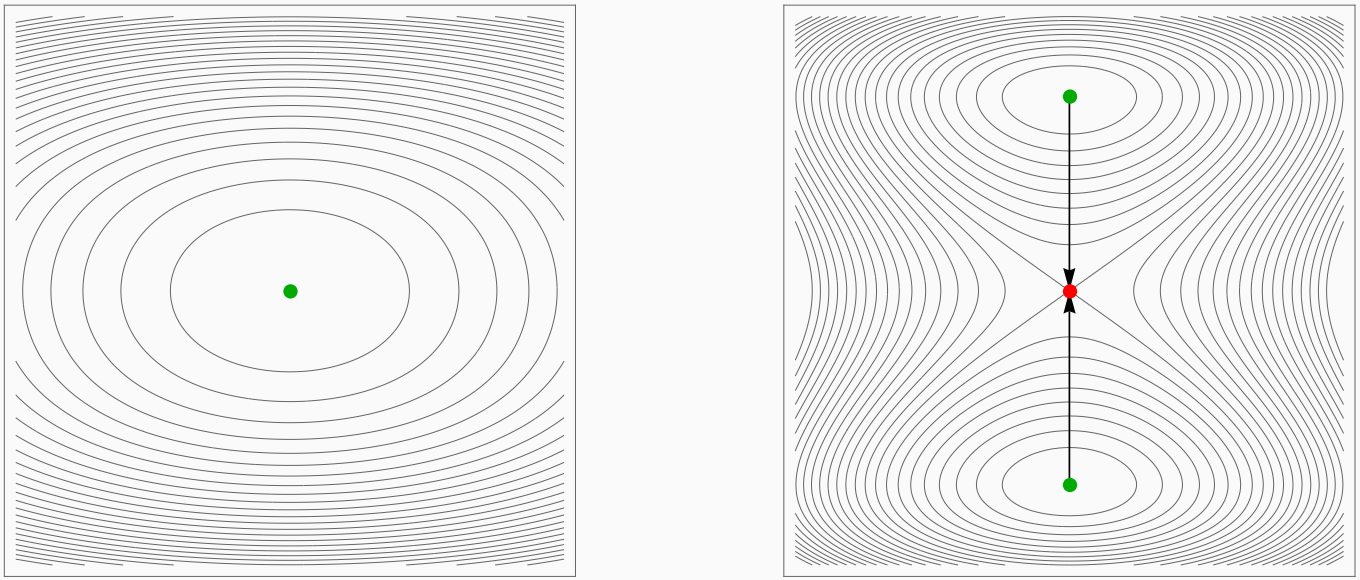


Figure 7: Period doubling bifurcation

Period doubling $(k = 2)$

- $\mu_{CZ}(\gamma_+^2; \gamma_-^2) = -1$ or 1. We assume -1 below.
- $\mu_{CZ}(\gamma_{new}; \gamma_-^2) = 0$.

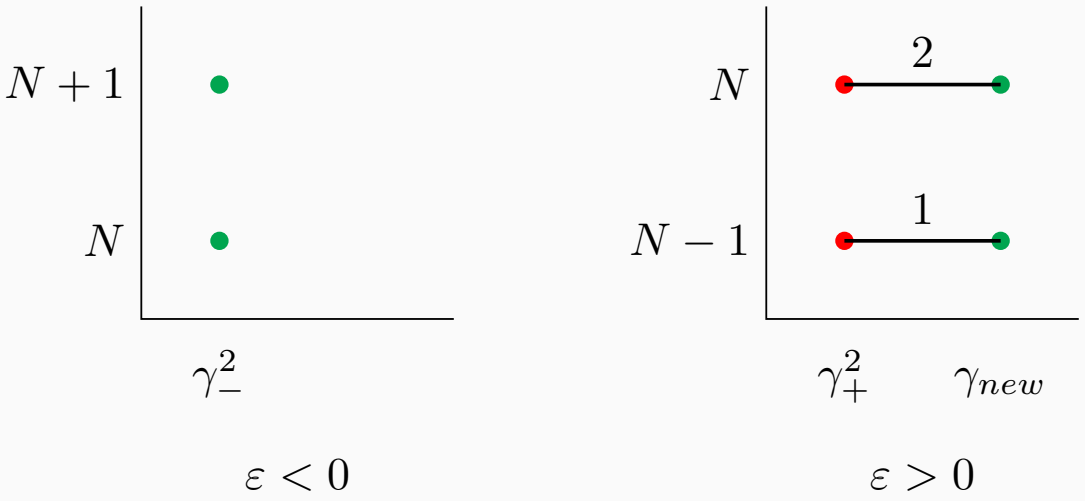


Figure 8: Chain complex mutation across the period doubling. $\check{\gamma}_H$ and $\check{\gamma}^2$ cancel each other, while $\hat{\gamma}_H$ and $\hat{\gamma}^2$ do not.

$$HF_*^{\mathrm{loc}}(\gamma_0^2) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^2, \\ 0 & \text{otherwise} \end{cases}$$

Research Plan

Symmetric Bifurcations and Lagrangian Floer theory

There are another types of bifurcations if the system has additional symmetries.

- CRTBP has one anti-symplectic involution and one symplectic involution, which makes the system D_2-symmetric.
- Hill's lunar problem has $\mathbb{Z}_2 \times D_2$-symmetry.
- Hénon–Heiles system has D_3-symmetry.

First example is the **pitchfork bifurcation**, which occurs for the simple orbit instead of the birth-death bifurcation.

Our previous work [JL26] covers some cases with $\mathbb{Z}_2$-symmetry, and we will extend this framework to broader classes of symmetric systems.

For anti-symplectic involution, Lagrangian Floer theory would also play a role: half-orbit is Lagrangian chord of the fixed locus.

Symmetry-breaking

Let there be an additional parameter μ such that $H_{\varepsilon,0}$ is G-symmetric but $H_{\varepsilon,\mu}$ is not for $\mu \neq 0$.

- At $\mu = 0$, symmetric bifurcation happens.
- At $\mu \neq 0$, this breaks into several generic bifurcations.

We expect that this would give some exact sequence between LFCCs, and provide an entrance to the multi-parameter bifurcations.

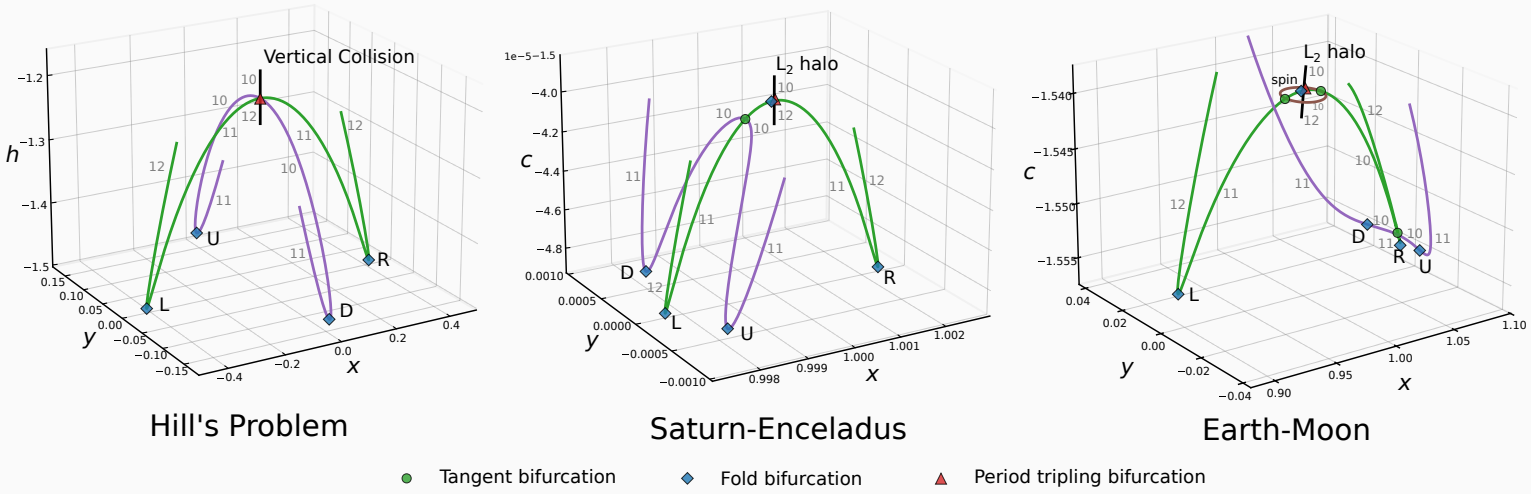


Figure 9: Symmetry-breaking, [JKvK26].

Persistence Modules and Floer–Cerf Theory

We can regard $CF_*(H_\varepsilon)$ as a 1-parameter family of chain complexes, suggesting a persistence-type structure.

Cerf theory describes how Morse complexes change along a parameter. What is the corresponding picture in Floer theory?

- We have the local ingredients: the classification of bifurcations and the corresponding changes of the LFCC.
- How do the moduli of non-local Floer cylinders change? Not clear.

Goal: assemble these local mutations into a global Floer-theoretic analogue of Cerf theory.

References i


Cengiz Aydin and Alexander Batkhin.
Studying network of symmetric periodic orbit families of the Hill problem via symplectic invariants.
Celestial Mech. Dynam. Astronom., 137(2):Paper No. 12, 77, 2025.


Michèle Audin and Mihai Damian.
Morse theory and Floer homology.
Universitext. Springer, London; EDP Sciences, Les Ulis, 2014.
Translated from the 2010 French original by Reinie Erné.


Cengiz Aydin, Urs Frauenfelder, Otto van Koert, Dayung Koh, and Agustin Moreno.
Symplectic geometry and space mission design.
2023.

References ii


Chankyu Joung, Dayung Koh, and Otto van Koert.
Bifurcations of highly inclined near halo orbits using moser regularization, 2026.


Hong-Kwon Jo and Dongho Lee.
Mutations of local floer chain complexes across hamiltonian bifurcations, 2026.


MaxwellMolecule.
K-s equation period doubling.
Wikimedia Commons, February 2021.
Illustration of period doubling in the Kuramoto–Sivashinsky equation. CC0 1.0 Universal Public Domain Dedication.

References iii



K. R. Meyer.

Generic bifurcation of periodic points.

Transactions of the American Mathematical Society, 149(1):95–107, 1970.



Juan-Pablo Ortega and Tudor Ratiu.

A dirichlet criterion for the stability of periodic and relative periodic orbits in hamiltonian systems.

Journal of Geometry and Physics, 32:131–159, 12 1999.



Figure 10: Thank you for your attention!

Questions, comments, and further discussion are welcome!